

EMILIA Protocol — the seatbelt for the agentic era

Verifiable human authorization for the irreversible actions AI agents are starting to take.

For fifty years, security answered "who is allowed in?" The dominant users of software are now AI agents that move money, change records, and delete data on the fly. The unanswered question is **authorization at the moment of action: who approved that exact action, before it executed?** "A human was in the loop" is, today, an unfalsifiable claim — and regulators (EU AI Act Art. 14) now require oversight no one can verify after the fact.

WHAT EP IS

Before a high-risk action runs, a **named human** approves the exact action on their own device (WebAuthn / Face ID). EP emits a signed **authorization receipt** anyone can verify **offline, no account, without trusting our servers** — and it stays valid even if EP disappears. *Decision logs are testimony; EP produces receipts.*

WHY IT'S REAL, AND CHECKABLE

- **3 languages / 8 suites:** JS, Python, Go verifiers agree on 8 adversarial cross-language conformance suites — the IETF bar for a standard.
- **Independently verified (2026):** an outside implementer ran the crash-test + conformance on their own machine and confirmed it — the first external reliance event.
- **Formal + standardized:** machine-checked TLA+ & Alloy in CI; authorization-receipt and quorum Internet-Drafts posted at the IETF.

THE POSITION — CONVERGENCE, NOT COMPETITION

A dozen-plus efforts now define "a receipt for an agent's action." EP authored the **Authorization Evidence Chain** that composes them into one offline decision — and supplies the one leg none of the others do: **a named human's authorization**. EP is the convergence point of the field.

WEDGE: HEALTHCARE

The two-person rule is **already mandated** there (the high-alert-med double-check). EP makes it verifiable, PHI-free, and Article-14-aligned. B2B2H via healthcare-AI vendors + procurement platforms.

MODEL: OPEN CORE

Format & verifier are free (ubiquity); the **operated trust root** — directory, transparency log, revocation, evidence pipelines — is the revenue layer an org can't casually self-host.

WHY INEVITABLE

Public-key crypto existed two decades before the browser padlock made its *absence* unacceptable. EP becomes obvious the same way — an insurer's coverage condition, an auditor's framework, or a government AI-auditor registry making the **absence of a receipt** unacceptable. Precedent → recommendation → requirement.

Open standard (Apache-2.0) · IETF Internet-Drafts · 3 verifiers / 8 conformance
suites · independently re-verified · TLA+ & Alloy in CI · NIST AI RMF + EU AI Act
mappings · try it offline: `npx @emilia-protocol/crash-test`

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